IEOR 221

Homework 10

Module 11 Portfolio Theory (November 12, 2025)

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1) The following are estimates for two stocks.

Stock	Expected Return	Beta	Firm-Specific Standard Deviation
A	13%	0.8	30%
B	18%	1.2	40%

The market index has a standard deviation of 22% and the risk-free rate is 8%.
a. What are the standard deviations of stocks A and B?
b. Suppose that we were to construct a portfolio with proportions: Stock A: 30%, Stock B: 45%, Risk Free Bond: 25%. Compute the expected return, standard deviation, beta, and nonsystematic standard deviation of the portfolio.

2) Assume that the risk-free rate of interest is 6% and the expected rate of return on the market is 16%.
a. A share of stock sells for $50 today. It will pay a dividend of $6 per share at the end of the year. Its beta is 1.2. What do investors expect the stock to sell for at the end of the year?
b. An investor is buying a firm with an expected perpetual annual cash flow of $100,000 but is unsure of its risk. If he thinks the beta of the firm is 0.5, when in fact the beta is really 1.0, how much more will he offer for the firm than it is truly worth?
c. A stock has an expected rate of return of 4%. What is its beta?

3)
a. A mutual fund with beta of 0.8 has an expected rate of return of 14%. If risk free rate r = 5%, and you expect the rate of return on the market portfolio to be 15%, should you invest in this fund?
b. What passive portfolio comprised of a market-index portfolio and a money market account would have the same beta as the fund?